AI/ML Engineer 5-Month Master Plan (22 Weeks)

Comprehensive Printable Roadmap | Nov 2, 2026 – Apr 4, 2027 | Featuring CampusX ML & CodeWithHarry Tracks

Resource Strategy & Channel Allocation:
• **Phase 1 (Python Core & OOP)**: CodeWithHarry 100 Days + Harvard CS50P for clean programming syntax and fundamentals.
• **Phase 2 (Math & Classical ML)**: **CampusX 100 Days of ML (Nitish Singh)** as the primary core engine for Gradient Descent, Regression, Trees, XGBoost, and Scikit-Learn Pipelines.
• **Phase 3 to 5 (Backend, RAG & Agents)**: FastAPI docs, Docker, CampusX RAG & LangGraph series + official Model Context Protocol (MCP) specifications.
• **Print & Track with a Pen**: Checkboxes [ ] and formatted pen-writing log boxes on every weekly page.

Master 22-Week Progress Tracker (Tick with Pen)

Status	Wk	Phase & Primary Source	Dates	Core Weekly Milestone & Deliverable
[ ]	1	P1: CodeWithHarry / CS50P	Nov 2 - 8	Number Guessing Game CLI & Exercises 1-10
[ ]	2	P1: CodeWithHarry / Git	Nov 9 - 15	Contact Book CLI with JSON storage on GitHub
[ ]	3	P1: CodeWithHarry / OOP	Nov 16 - 22	OOP Refactored Contact Book with Custom Exceptions
[ ]	4	P1: SQL & Kaggle	Nov 23 - 29	E-Commerce Schema + 15 Analytical SQL Queries
[ ]	5	P1: CampusX NumPy/Pandas	Nov 30 - Dec 6	Kaggle Telco Churn Data Profiling & Cleaning
[ ]	6	P2: CampusX Math & GD	Dec 7 - 13	Gradient Descent from scratch in pure NumPy
[ ]	7	P2: CampusX Regression	Dec 14 - 20	Linear & Logistic Regression + Evaluation Metrics
[ ]	8	P2: CampusX Trees/XGB	Dec 21 - 27	Random Forest vs. XGBoost Benchmark on Churn
[ ]	9	P2: CampusX Pipelines	Dec 28 - Jan 3	Scikit-Learn Pipeline + ColumnTransformer (No Leakage)
[ ]	10	P2: Portfolio Project 1	Jan 4 - 10	**PORTFOLIO 1: End-to-End Churn ML Engine on GitHub**
[ ]	11	P3: CampusX / FastAPI	Jan 11 - 17	Model Prediction REST API with Pydantic Validation
[ ]	12	P3: DB & SQLAlchemy	Jan 18 - 24	PostgreSQL Prediction Logging with Dependency Injection
[ ]	13	P3: Auth & Redis	Jan 25 - 31	JWT Auth + Redis Cache-Aside for Model Endpoints
[ ]	14	P3: Portfolio Project 2	Feb 1 - 7	**PORTFOLIO 2: Containerized ML Inference Microservice**
[ ]	15	P4: PyTorch Deep Learning	Feb 8 - 14	Multi-Layer Perceptron (MLP) trained from scratch
[ ]	16	P4: Transformers & Text	Feb 15 - 21	Semantic Search CLI with Sentence Transformers
[ ]	17	P4: CampusX Vector DBs	Feb 22 - 28	Document Ingestion & Indexing Pipeline (pgvector/Chroma)
[ ]	18	P4: CampusX RAG Engine	Mar 1 - 7	**PORTFOLIO 3: Enterprise RAG Knowledge Engine API**
[ ]	19	P5: CampusX LangGraph	Mar 8 - 14	LangGraph Agent executing multi-step DB & Web tools
[ ]	20	P5: Model Context Protocol	Mar 15 - 21	Custom MCP Server connecting internal tools to AI clients
[ ]	21	P5: Capstone Platform	Mar 22 - 28	**FLAGSHIP CAPSTONE: AI Support & Knowledge Platform**
[ ]	22	P5: Career & Interviews	Mar 29 - Apr 4	**Resume, GitHub Audit, Mock Interviews & Job Sprint**

Week 1: Python Core Essentials, Input/Output & Conditionals

Phase: Phase 1: Software Engineering Core (Diwali Week) | **Dates**: Nov 2 – Nov 8, 2026 (Diwali on Nov 8; Nov 6-8 Light Festive Buffer)
Primary Goal: Write clean Python programs with variables, data types, user input, math operators, conditionals, and loops.
English Resource: Harvard CS50P: Week 0 (Functions/Variables) & Week 1 (Conditionals) | **Math**: Khan Academy India (Hindi): Fractions, Decimals, Order of Operations (www.khanacademy.org/math/hindi)
Notes & Guides: CodeWithHarry Python Ch 1-2 Notes & Harvard CS50P Notes Week 0-1
Weekly Milestone: **Build Number Guessing Game CLI & Complete Exercises 1–10 from Starter Pack**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	Day 1	**Day #1: Intro to Programming & Python [CWH]**	What is programming, interpreter vs compiler, Python 3 installation, VS Code setup.	Watch Video
[ ]	Day 1	**Day #2: Some Amazing Python Programs [CWH]**	Ecosystem overview: AI, ML, Web APIs, Automation. What Python can do.	Watch Video
[ ]	Day 1	**Day #3: Modules and Pip in Python [CWH]**	Built-in modules vs external modules, pip install, terminal commands.	Watch Video
[ ]	Day 1	**Day #4: Our First Python Program [CWH]**	print('Hello World'), code execution flow, indentation rules.	Watch Video
[ ]	Day 2	**Day #5: Comments & Escape Sequences [CWH]**	Single/multi-line comments, \n, \t, print sep=' ' and end='\n' parameters.	Watch Video
[ ]	Day 2	**Day #6: Variables & Data Types [CWH]**	Variables, int, float, str, bool, complex, type() function, type inspection.	Watch Video
[ ]	Day 2	**Day #7 & #8: Operators & Calc Project [CWH]**	Arithmetic operators (+, -, *, /, //, %, **). Build basic calculator CLI.	Watch Video
[ ]	Day 3	**Day #9: Typecasting in Python [CWH]**	Explicit vs implicit type conversion, converting str to int/float.	Watch Video
[ ]	Day 3	**Day #10: Taking User Input [CWH]**	input() function, handling user text and casting to numbers immediately.	Watch Video
[ ]	Day 4	**Day #14: If Else Conditionals [CWH]**	Comparison operators (==, !=, >, <, >=, <=), if, elif, else indentation logic.	Watch Video
[ ]	Day 4	**Day #17: For Loops in Python [CWH]**	for item in sequence, iterating strings, range(start, stop, step) function.	Watch Video
[ ]	Day 5	**Day #18: While Loops in Python [CWH]**	while condition, loop counters, incrementing, avoiding infinite loops.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 2: Data Structures, Functions & Git Fundamentals

Phase: Phase 1: Software Engineering Core (Post-Diwali Week) | **Dates**: Nov 9 – Nov 15, 2026 (Bhai Dooj on Nov 10; Weekend Wedding Buffer)
Primary Goal: Master Python data collections (Lists, Tuples, Dictionaries, Sets), reusable functions, file handling, and Git version control.
English Resource: Harvard CS50P: Week 2 (Loops & Collections) & Week 6 (File I/O) | **Math**: Khan Academy: Coordinate Planes, Graphing Equations & Linear Expressions
Notes & Guides: CodeWithHarry Notes Ch 3, 4, 5, 8 & Pro Git Book (git-scm.com/book/en/v2)
Weekly Milestone: **Contact Book CLI with JSON storage (Add, List, Search, Delete) committed to GitHub**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 8**	**Day #11 & #12: Strings & Slicing [CWH]**	String indexing, negative indexing, slice notation [start:stop:step], len().	Watch Video
[ ]	**Day 8**	**Day #13: String Methods in Python [CWH]**	upper(), lower(), strip(), replace(), split(), count(), find(), startswith().	Watch Video
[ ]	**Day 9**	**Day #20 & #21: Functions & Arguments [CWH]**	def, parameters vs arguments, default values, return statement, *args/**kwargs.	Watch Video
[ ]	**Day 10**	**Day #22 & #23: Lists & List Methods [CWH]**	List indexing, mutable behavior, append(), insert(), extend(), pop(), sort().	Watch Video
[ ]	**Day 11**	**Day #24 & #25: Tuples in Python [CWH]**	Tuple immutability, tuple methods, packing/unpacking, converting to list.	Watch Video
[ ]	**Day 11**	**Day #31 & #32: Sets & Set Operations [CWH]**	Unordered unique elements, union(), intersection(), difference(), symmetric_diff.	Watch Video
[ ]	**Day 12**	**Day #33 & #34: Dictionaries & Methods [CWH]**	Key-value mappings, dict[key] vs dict.get(), keys(), values(), items(), update().	Watch Video
[ ]	**Day 13**	**Git & GitHub Crash Course [Chai aur Code]**	git init, git add, git commit, git status, git log, creating repo on GitHub, push.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 3: Object-Oriented Programming (OOP) & Practical Python

Phase: Phase 1: Software Engineering Core (Wedding Season Buffer) | **Dates**: Nov 16 – Nov 22, 2026 (Wedding / Reception Weekend Buffer)
Primary Goal: Write modular, professional code using Classes, Constructors, Inheritance, Exception Handling, and File I/O.
English Resource: Harvard CS50P: Week 8 (Object-Oriented Programming) | **Math**: Khan Academy: Statistical Measures (Mean, Median, Mode, Variance Intuition)
Notes & Guides: CodeWithHarry Notes Ch 9, 10, 11 & CampusX Python OOP Guides
Weekly Milestone: **Refactor Contact Book into clean OOP classes with Custom Exceptions and Logging**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 15**	**Day #36 & #37: Exception Handling [CWH]**	try, except, else, finally blocks, ValueError, ZeroDivisionError handling.	Watch Video
[ ]	**Day 15**	**Day #38: Raising Custom Errors [CWH]**	raise keyword, inheriting from Exception, defining meaningful error messages.	Watch Video
[ ]	**Day 16**	**Day #49 & #50: File I/O in Python [CWH]**	open(filename, mode), 'r', 'w', 'a', with open(...) as f context managers, readlines().	Watch Video
[ ]	**Day 17**	**OOPs in Python: Classes & Objects [CampusX / CWH]**	Classes as blueprints, objects as instances, attributes, self parameter.	Watch Video
[ ]	**Day 17**	**Constructors & Instance State [CampusX]**	__init__ dunder method, initializing object state, instance attributes.	Watch Video
[ ]	**Day 18**	**Encapsulation, Getters & Setters [CampusX]**	@property decorator, encapsulation, protecting internal state.	Watch Video
[ ]	**Day 19**	**Inheritance & Polymorphism [CampusX]**	Single inheritance, child class inheriting parent methods, code reuse.	Watch Video
[ ]	**Day 19**	**Dunder Magic Methods (__str__, __repr__) [CampusX]**	super() call, __str__, __repr__, __len__ magic methods.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 4: Relational Databases & SQL for Data/ML

Phase: Phase 1: Database & Data Architecture | **Dates**: Nov 23 – Nov 29, 2026 (Event / Social Buffer Day Included)
Primary Goal: Model data relationships, create schemas, and write complex analytical queries with joins and aggregations.
English Resource: Kaggle: Intro to SQL & Advanced SQL Courses | **Math**: Khan Academy: Set Theory, Venn Diagrams & Relational Logic
Notes & Guides: Kaggle SQL Tutorials & PostgreSQL Official Tutorial Docs
Weekly Milestone: **E-Commerce Database Schema (Users, Products, Orders) + 15 Analytical Queries**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 22**	**SQL Foundations: Databases & Tables [CampusX SQL]**	What is RDBMS, SQLite vs PostgreSQL, tables, rows, columns, data types.	Watch Video
[ ]	**Day 22**	**SQL CRUD: CREATE, INSERT, SELECT [CampusX SQL]**	CREATE TABLE, Primary Keys, INSERT INTO, basic SELECT * FROM table.	Watch Video
[ ]	**Day 23**	**Filtering & Sorting: WHERE, ORDER BY [CampusX SQL]**	WHERE clause, comparison operators, AND, OR, NOT, IN, BETWEEN, LIKE, LIMIT.	Watch Video
[ ]	**Day 24**	**Aggregations: GROUP BY & HAVING [CampusX SQL]**	COUNT(), SUM(), AVG(), MIN(), MAX(), GROUP BY columns, filtering with HAVING.	Watch Video
[ ]	**Day 25**	**Relational Modeling & JOINS [CampusX SQL]**	Foreign keys, 1-to-many relationships, INNER JOIN, LEFT JOIN, FULL OUTER JOIN.	Watch Video
[ ]	**Day 26**	**Subqueries & Window Functions [CampusX SQL]**	Nested SELECT subqueries, Common Table Expressions (WITH CTE), ROW_NUMBER().	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 5: NumPy & Pandas Data Wrangling (CampusX)

Phase: Phase 1: Numerical Data Wrangling | **Dates**: Nov 30 – Dec 6, 2026 (Catch-up / Wedding Buffer Day Included)
Primary Goal: Master numerical arrays and tabular data using CampusX's practical data manipulation series on Kaggle datasets.
English Resource: FreeCodeCamp: Data Analysis with Python (NumPy & Pandas Full Course) | **Math**: Khan Academy: Matrix Addition, Multiplication, Dot Products & Summary Statistics
Notes & Guides: CampusX NumPy/Pandas Notebooks & Pandas Official Guide
Weekly Milestone: **Clean and profile a Kaggle dataset (e.g. Telco Churn CSV) with summary tables**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 29**	**What is NumPy & Array Creation [CampusX ML]**	np.array(), ndarray attributes (.shape, .dtype, .ndim), np.zeros, np.ones, np.arange.	Watch Video
[ ]	**Day 29**	**Vectorization & Array Broadcasting [CampusX ML]**	Element-wise math (+, -, *, /), dot products (@), broadcasting rules, memory speed.	Watch Video
[ ]	**Day 30**	**NumPy Indexing, Slicing & Masking [CampusX ML]**	2D slicing arr[row, col], boolean masking arr[arr > 5], filtering arrays without loops.	Watch Video
[ ]	**Day 31**	**Pandas Series & DataFrames [CampusX ML]**	pd.Series, pd.DataFrame, reading CSV files (pd.read_csv), .head(), .info(), .describe().	Watch Video
[ ]	**Day 32**	**Handling Missing Values & Outliers [CampusX ML]**	Handling nulls (.isna(), .dropna(), .fillna()), mean/median imputation, outlier checks.	Watch Video
[ ]	**Day 33**	**Filtering & GroupBy Aggregations [CampusX ML]**	Boolean indexing df[df['age'] > 25], .groupby('category').agg(), .pivot_table().	Watch Video
[ ]	**Day 34**	**Merging & Joining DataFrames [CampusX ML]**	pd.merge() for SQL-like joins (inner, left), pd.concat() for row/column stacking.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 6: Gradient Descent & Math for ML (CampusX Core)

Phase: Phase 2: Math & Classical Machine Learning | **Dates**: Dec 7 – Dec 13, 2026 (Wedding / Reception Weekend Buffer)
Primary Goal: Build mathematical and visual intuition for Gradient Descent from scratch using CampusX's dedicated lectures.
English Resource: 3Blue1Brown: Essence of Linear Algebra & Essence of Calculus | **Math**: Khan Academy India: Probability Distributions, Normal Distribution & Derivatives
Notes & Guides: CampusX 100 Days of ML: Gradient Descent & Math Notes
Weekly Milestone: **Implement Gradient Descent in pure NumPy for a line y = mx + b; plot the loss curve**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 36**	**What is Machine Learning? [CampusX #1]**	AI vs ML vs DL, model formulation, loss minimization overview.	Watch Video
[ ]	**Day 37**	**Dot Products, Norms & Cosine Similarity [3B1B]**	Dot product intuition, L1 vs L2 norms, measuring angle/similarity between vectors.	Watch Video
[ ]	**Day 38**	**Derivatives & The Gradient Vector [3B1B]**	Partial derivatives, what the gradient vector points to, intuition of slope descent.	Watch Video
[ ]	**Day 39**	**Gradient Descent From Scratch [CampusX #47]**	Mathematical derivation, loss function derivative, weight update rule: w = w - lr * grad.	Watch Video
[ ]	**Day 40**	**Batch vs Stochastic vs Mini-Batch GD [CampusX #48]**	Comparing GD variants on convergence speed, computational overhead, and noisy steps.	Watch Video
[ ]	**Day 41**	**Loss Functions: MSE vs. Log Loss [CampusX]**	Mean Squared Error for regression, why MSE fails on classification, Binary Cross-Entropy.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 7: Linear & Logistic Regression (CampusX ML)

Phase: Phase 2: Math & Classical Machine Learning | **Dates**: Dec 14 – Dec 20, 2026 (Mid-December Wedding Buffer)
Primary Goal: Master linear models, Ridge/Lasso regularization, logistic decision boundaries, and evaluation metrics with CampusX.
English Resource: StatQuest with Josh Starmer: Machine Learning Fundamentals | **Math**: Khan Academy: Linear Equations, Best-Fit Lines & Sigmoid Activation Function
Notes & Guides: CampusX Regression Notes & Scikit-Learn Official User Guide
Weekly Milestone: **Train Logistic Regression model on a medical dataset; generate Confusion Matrix & PR curve**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 43**	**Simple Linear Regression Math & Code [CampusX #45]**	OLS closed-form solution, slope & intercept formula, fitting line on data.	Watch Video
[ ]	**Day 44**	**Multiple Linear Regression [CampusX #46]**	N-dimensional regression, matrix formulation (X^T X)^-1 X^T y, R2 score & adjusted R2.	Watch Video
[ ]	**Day 45**	**Ridge Regression (L2 Penalty) [CampusX #51]**	Overfitting prevention, coefficient shrinkage, lambda penalty geometric intuition.	Watch Video
[ ]	**Day 45**	**Lasso Regression & Feature Selection [CampusX #52]**	L1 penalty forcing weights to exact zero; sparsity and automatic feature selection.	Watch Video
[ ]	**Day 46**	**Logistic Regression Math & Sigmoid [CampusX #57]**	Odds ratio, log-odds, sigmoid mapping to (0, 1), maximum likelihood estimation.	Watch Video
[ ]	**Day 47**	**Confusion Matrix, Precision & Recall [CampusX #59]**	True Positive, False Positive, Precision vs Recall trade-off, F1-Score, ROC-AUC.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 8: Decision Trees, Random Forests & XGBoost (CampusX)

Phase: Phase 2: Math & Classical Machine Learning | **Dates**: Dec 21 – Dec 27, 2026 (Christmas Holiday & Social Buffer)
Primary Goal: Master non-linear decision trees, bagging ensembles, and gradient boosting algorithms with CampusX.
English Resource: StatQuest: Decision Trees, Random Forests & XGBoost Series | **Math**: Khan Academy: Logarithms, Entropy & Information Theory Basics
Notes & Guides: CampusX Ensemble Learning Notebooks & XGBoost Documentation
Weekly Milestone: **Benchmark Random Forest vs. XGBoost on Telco Churn dataset with feature importance charts**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 50**	**Decision Trees: Gini Impurity & Entropy [CampusX]**	How trees split data, calculating Gini impurity by hand, information gain, tree depth.	Watch Video
[ ]	**Day 51**	**Tree Pruning & Hyperparameters [CampusX]**	max_depth, min_samples_split, min_samples_leaf, preventing tree overfitting.	Watch Video
[ ]	**Day 52**	**Random Forests & Bagging Ensembles [CampusX]**	Bootstrap aggregating, row and column sampling, OOB score, voting classifier.	Watch Video
[ ]	**Day 53**	**AdaBoost & Sequential Learning [CampusX]**	Why boosting builds on previous errors, sample weights update mechanics.	Watch Video
[ ]	**Day 54**	**Gradient Boosting & XGBoost [CampusX]**	Fitting decision trees to pseudo-residuals, learning rate shrinkage, XGBoost speed.	Watch Video
[ ]	**Day 55**	**Feature Importance & Model Evaluation [CampusX]**	MDI feature importance, permutation importance, intro to SHAP values.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 9: Feature Engineering & Scikit-Learn Pipelines (CampusX)

Phase: Phase 2: Math & Classical Machine Learning | **Dates**: Dec 28, 2026 – Jan 3, 2027 (New Year Celebration Buffer)
Primary Goal: Build robust, reproducible pipelines using CampusX's dedicated ColumnTransformer and Pipeline masterclasses.
English Resource: Scikit-Learn Official Tutorial: Pipelines and Composite Estimators | **Math**: Khan Academy: K-Fold Cross Validation & Combinatorics of Hyperparameter Grids
Notes & Guides: CampusX Pipeline Notebooks & Scikit-Learn Common Pitfalls Guide
Weekly Milestone: **Build an end-to-end Pipeline with ColumnTransformer, OneHotEncoder & Scaler; save with joblib**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 57**	**Handling Categorical Data [CampusX #28]**	Ordinal encoding vs One-Hot Encoding, dummy variable trap, handle_unknown='ignore'.	Watch Video
[ ]	**Day 58**	**Feature Scaling: Standardization vs Norm [CampusX #27]**	StandardScaler (z-score) vs MinMaxScaler, when to use which on linear vs tree models.	Watch Video
[ ]	**Day 59**	**Scikit-Learn ColumnTransformer [CampusX #31]**	Applying different transformations to numeric vs categorical columns cleanly in one step.	Watch Video
[ ]	**Day 60**	**Building Scikit-Learn Pipelines [CampusX #32]**	Chaining preprocessing with estimator; preventing data leakage completely.	Watch Video
[ ]	**Day 61**	**Cross-Validation Done Right [CampusX]**	Stratified K-Fold inside pipelines, evaluating baseline without test data leakage.	Watch Video
[ ]	**Day 62**	**Pipeline Serialization with Joblib [CampusX]**	joblib.dump() and joblib.load(), testing predictions on unseen single-row inputs.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 10: PORTFOLIO PROJECT 1 — End-to-End ML Prediction Engine

Phase: Phase 2: Milestone Portfolio Project | **Dates**: Jan 4 – Jan 10, 2027 (New Year Sprint Week)
Primary Goal: Publish your first full-fledged Machine Learning portfolio project on GitHub following CampusX project best practices.
English Resource: Google ML Guides: Rules of Machine Learning (Best Practices for ML Engineering) | **Math**: Model Metric Synthesis: Precision-Recall Tradeoff curves & Business Cost Matrices
Notes & Guides: CampusX End-to-End Project Architecture & Model Cards
Weekly Milestone: **PORTFOLIO PROJECT 1 LIVE: End-to-End Customer Churn Prediction Engine on GitHub**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 64**	**End-to-End ML Project Walkthrough [CampusX]**	Problem framing, choosing dataset, building baseline model.	Watch Video
[ ]	**Day 65**	**Data Cleaning & Feature Engineering [CampusX]**	Feature selection, handling missing values, creating clean train/test splits.	Watch Video
[ ]	**Day 66**	**Pipeline Construction & Model Comparison [CampusX]**	Comparing Logistic Regression, Random Forest, and XGBoost on Stratified 5-Fold.	Watch Video
[ ]	**Day 67**	**Tuning & Final Model Evaluation [CampusX]**	Hyperparameter optimization, final test set evaluation, generating confusion matrix.	Watch Video
[ ]	**Day 68**	**Pipeline Serialization & Smoke Testing**	Packaging pipeline into model.joblib, writing sample prediction test script.	Watch Video
[ ]	**Day 69**	**Writing a Production-Grade README**	Documenting problem, metrics table, setup instructions, limitations, and future steps.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 11: HTTP, REST & FastAPI for Machine Learning

Phase: Phase 3: Production Backend & Model Serving | **Dates**: Jan 11 – Jan 17, 2027 (Winter Wedding Weekend Buffer)
Primary Goal: Build async APIs with FastAPI & Pydantic to serve Machine Learning predictions with interactive Swagger UI.
English Resource: FastAPI Official Tutorial: First Steps, Path Params, Query Params, Request Body | **Math**: Network Latency & Throughput Intuition (Requests Per Second - RPS, p95/p99 Latency)
Notes & Guides: FastAPI Official Documentation & CampusX API Deployment Guides
Weekly Milestone: **Build Model Prediction REST API with Pydantic validation returning dummy predictions**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 71**	**How the Web Works: HTTP, DNS & REST**	Client-server architecture, HTTP methods (GET, POST, PUT, DELETE), status codes.	Watch Video
[ ]	**Day 72**	**Deploying ML Models with FastAPI [CampusX / Tiangolo]**	Why FastAPI for ML, async/await in Python, installing uvicorn, running your first @app.get('/').	Watch Video
[ ]	**Day 73**	**Pydantic v2: Schema Validation & Typing**	BaseModel, input feature validation, automatic 422 Unprocessable Entity error handling.	Watch Video
[ ]	**Day 74**	**Path & Query Parameters in FastAPI**	Path(..., ge=1), Query filtering, optional parameters, returning structured JSON responses.	Watch Video
[ ]	**Day 75**	**Error Handling & HTTP Exceptions**	Raising HTTPException(status_code=404), custom exception handlers in FastAPI.	Watch Video
[ ]	**Day 76**	**Interactive API Documentation**	Exploring Swagger UI (/docs) and ReDoc (/redoc), testing endpoints in browser.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 12: Database Integration & SQLAlchemy 2.0 ORM

Phase: Phase 3: Production Backend & Model Serving | **Dates**: Jan 18 – Jan 24, 2027 (Wedding / Family Event Buffer)
Primary Goal: Persist data with SQLAlchemy 2.0 and PostgreSQL; inject database sessions cleanly using FastAPI Dependencies.
English Resource: FastAPI Documentation: SQL (Relational) Databases with SQLAlchemy | **Math**: Connection Pool Sizing: Concurrency, Thread Pools & Database Contention
Notes & Guides: SQLAlchemy 2.0 Unified Tutorial & Migration Guidelines
Weekly Milestone: **Connect Prediction API to PostgreSQL to log every prediction request, latency, and inputs**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 78**	**SQLAlchemy 2.0 Declarative Models**	DeclarativeBase, Mapped types, mapped_column, primary keys, relationships.	Watch Video
[ ]	**Day 79**	**Engine, Sessionmaker & Connection Pooling**	create_engine, SessionLocal, pool_size, max_overflow, managing DB lifecycles.	Watch Video
[ ]	**Day 80**	**Repository Pattern & CRUD Operations**	Separating DB queries from API routes: db.execute(select(...)), insert, commit.	Watch Video
[ ]	**Day 81**	**FastAPI Dependency Injection (Depends)**	yield get_db(), ensuring sessions close cleanly after each request completes.	Watch Video
[ ]	**Day 82**	**Database Migrations with Alembic**	alembic init, env.py setup, alembic revision --autogenerate, alembic upgrade head.	Watch Video
[ ]	**Day 83**	**Building the Prediction Logger Entity**	Storing prediction_id, user_id, input_features, prediction, latency_ms, created_at.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 13: Authentication, Security & Redis Caching

Phase: Phase 3: Production Backend & Model Serving | **Dates**: Jan 25 – Jan 31, 2027 (End-of-Month Review & Buffer)
Primary Goal: Protect API endpoints with JWT authentication and implement Redis caching to eliminate redundant computations.
English Resource: FastAPI Tutorial: Security with OAuth2 and Password Hashing | **Math**: Hash Functions & Salt Mechanics (Bcrypt Work Factor & Expiration Time Math)
Notes & Guides: FastAPI Security & JWT Docs; Redis Developer Quickstart
Weekly Milestone: **Secured API with /register, /login, JWT protected /predict route, and Redis prediction caching**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 85**	**Password Hashing with Passlib & Bcrypt**	Never store plaintext; salts, one-way hash functions, pwd_context.hash/verify.	Watch Video
[ ]	**Day 86**	**JSON Web Tokens (JWT) Deep Dive**	Header, Payload, Signature; issuing tokens, setting expiration times (exp).	Watch Video
[ ]	**Day 87**	**OAuth2PasswordBearer & Protected Routes**	FastAPI OAuth2 flow, extracting bearer token, decoding JWT, getting current user.	Watch Video
[ ]	**Day 88**	**Redis Foundations: In-Memory Data Store**	Why Redis is blazing fast, key-value data structures, TTL (Time To Live).	Watch Video
[ ]	**Day 89**	**Cache-Aside Pattern for ML Inference**	Hashing input features into cache key; returning cached prediction if hit.	Watch Video
[ ]	**Day 90**	**API Rate Limiting with Redis**	Sliding window rate limiting with Redis to protect costly endpoints from abuse.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 14: PORTFOLIO PROJECT 2 — Containerized ML Inference Microservice

Phase: Phase 3: Milestone Portfolio Project | **Dates**: Feb 1 – Feb 7, 2027 (Docker & Production Sprint)
Primary Goal: Package your trained ML model, FastAPI backend, PostgreSQL, and Redis into Docker containers and deploy.
English Resource: Docker Documentation: Multi-stage Builds & Compose for Python Microservices | **Math**: Container Resource Allocation: CPU Quotas, Memory Limits & OOM Killers
Notes & Guides: Docker Official Get Started Guide & Compose Specification
Weekly Milestone: **PORTFOLIO PROJECT 2 LIVE: Containerized ML Inference Microservice with Docker Compose**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 92**	**Containers vs Virtual Machines**	Kernel namespaces, cgroups, why containers are lightweight, Docker architecture.	Watch Video
[ ]	**Day 93**	**Writing a Production Dockerfile**	Multi-stage builds, non-root user, slim Python base, .dockerignore best practices.	Watch Video
[ ]	**Day 94**	**Multi-Container Docker Compose**	Orchestrating web, db (Postgres), and cache (Redis) with health checks & networks.	Watch Video
[ ]	**Day 95**	**Loading ML Models on Startup (Lifespan)**	FastAPI lifespan async context manager; loading joblib pipeline into memory once.	Watch Video
[ ]	**Day 96**	**Automated Testing with Pytest & HTTPX**	Unit testing routes, testing auth failures, testing model output schema format.	Watch Video
[ ]	**Day 97**	**Free Cloud Deployment (Railway / Render)**	Pushing Docker image, configuring environment variables, verifying public live URL.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 15: PyTorch & Neural Network Foundations

Phase: Phase 4: Deep Learning, Embeddings & RAG | **Dates**: Feb 8 – Feb 14, 2027 (Valentine's / Event Buffer Weekend)
Primary Goal: Build deep learning intuition; master PyTorch tensors, autograd, custom layers, and the training loop.
English Resource: PyTorch Official Tutorials: Quickstart & Deep Learning with PyTorch: A 60 Minute Blitz | **Math**: Khan Academy / 3Blue1Brown: What is a Neural Network & Gradient Backpropagation
Notes & Guides: PyTorch Official Tutorials: Learn the Basics (Tensors to Training Loops)
Weekly Milestone: **Train a 3-Layer MLP from scratch in PyTorch on tabular data; log training loss curves**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 99**	**PyTorch Tensors & Hardware Acceleration**	Tensors vs NumPy arrays, shapes, slicing, moving tensors to GPU (cuda/mps).	Watch Video
[ ]	**Day 100**	**Autograd: Automatic Differentiation**	Computational graph, requires_grad=True, loss.backward(), tensor.grad accumulation.	Watch Video
[ ]	**Day 101**	**Building Neural Networks with nn.Module**	nn.Linear, activation functions (ReLU, GELU), defining forward() method.	Watch Video
[ ]	**Day 102**	**Loss Functions & Optimizers in PyTorch**	CrossEntropyLoss, MSELoss, AdamW optimizer, optimizer.zero_grad(), optimizer.step().	Watch Video
[ ]	**Day 103**	**PyTorch Dataset & DataLoader**	Custom __len__ and __getitem__, batch_size, shuffle=True, multi-worker loading.	Watch Video
[ ]	**Day 104**	**The Canonical PyTorch Training Loop**	Epoch loop, train mode vs eval mode (model.train(), model.eval(), torch.no_grad()).	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 16: Transformers, Tokenization & Text Embeddings

Phase: Phase 4: Deep Learning, Embeddings & RAG | **Dates**: Feb 15 – Feb 21, 2027 (Mid-Feb Event Buffer)
Primary Goal: Master the Transformer architecture (Self-Attention), tokenizers, and dense semantic vector embeddings.
English Resource: Hugging Face NLP Course: Transformers, Tokenizers & Pipeline API | **Math**: Self-Attention Math: Query, Key, Value Dot Products & Softmax Normalization
Notes & Guides: Hugging Face NLP Course: Transformer Models & CampusX GenAI Guides
Weekly Milestone: **Build a Semantic Search CLI using sentence-transformers to find top-3 closest documents**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 106**	**Evolution of NLP: TF-IDF to Transformers**	Limitations of bag-of-words and RNNs; why parallel attention conquered NLP.	Watch Video
[ ]	**Day 107**	**The Transformer: Self-Attention Explained**	Query, Key, Value intuition; scaled dot-product attention formula; multi-head attention.	Watch Video
[ ]	**Day 108**	**Encoder vs Decoder Architectures**	BERT (bi-directional encoder for embeddings) vs GPT (autoregressive decoder for generation).	Watch Video
[ ]	**Day 109**	**Tokenization Mechanics (BPE & WordPiece)**	Subword tokenization, vocabulary files, token IDs, context window limitations.	Watch Video
[ ]	**Day 110**	**Dense Vector Embeddings in Practice [CampusX GenAI]**	Mapping sentences to 384/1536-dimensional vectors; cosine distance between concepts.	Watch Video
[ ]	**Day 111**	**HuggingFace sentence-transformers**	Loading all-MiniLM-L6-v2, model.encode(sentences), calculating similarity matrices.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 17: Vector Databases & Document Chunking (CampusX)

Phase: Phase 4: Deep Learning, Embeddings & RAG | **Dates**: Feb 22 – Feb 28, 2027 (End-of-Feb Buffer)
Primary Goal: Store, index, and retrieve millions of vectors efficiently with HNSW indexes and intelligent chunking.
English Resource: Pinecone / ChromaDB Guides: Chunking Strategies for LLM Applications | **Math**: Hierarchical Navigable Small World (HNSW) Graphs & Approximate Nearest Neighbor Math
Notes & Guides: ChromaDB Documentation & CampusX Vector Store Architecture Guide
Weekly Milestone: **Ingestion Pipeline that parses PDFs/markdown files, chunks them, and stores embeddings with metadata**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 113**	**Vector Databases vs Relational Databases [CampusX RAG]**	Exact nearest neighbor (O(N) scan) vs Approximate Nearest Neighbor (ANN) index.	Watch Video
[ ]	**Day 114**	**HNSW & Vector Indexing Under the Hood**	Multi-layer skip graphs for vectors, distance metrics (Cosine vs Euclidean vs Dot product).	Watch Video
[ ]	**Day 115**	**Setting up ChromaDB & pgvector [CampusX]**	Running ChromaDB in Docker, enabling vector extension in PostgreSQL (CREATE EXTENSION vector).	Watch Video
[ ]	**Day 116**	**Document Chunking Strategies [CampusX]**	Fixed-size vs recursive character chunking; chunk_size vs chunk_overlap trade-offs.	Watch Video
[ ]	**Day 117**	**Metadata Filtering During Retrieval**	Pre-filtering vs post-filtering by document category, author, date, or user permissions.	Watch Video
[ ]	**Day 118**	**Hybrid Search: BM25 + Vector Search**	Why vector search fails on exact keywords/part numbers; combining lexical with semantic.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 18: PORTFOLIO PROJECT 3 — Enterprise RAG Knowledge Engine API

Phase: Phase 4: Milestone Portfolio Project | **Dates**: Mar 1 – Mar 7, 2027 (RAG Sprint Week)
Primary Goal: Build and ship an end-to-end Retrieval-Augmented Generation (RAG) backend API with streaming citations.
English Resource: Google Gemini API Guides: RAG & Grounding with Custom Data | **Math**: RAG Evaluation Metrics: Faithfulness, Answer Relevance & Context Recall Formulas
Notes & Guides: CampusX Complete RAG Guide & OpenAI/Gemini Architecture Patterns
Weekly Milestone: **PORTFOLIO PROJECT 3 LIVE: Enterprise RAG Knowledge Engine API with token streaming**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 120**	**The Production RAG Architecture [CampusX RAG]**	Query rewording, dense retrieval, context window prompt injection, hallucination control.	Watch Video
[ ]	**Day 121**	**Calling LLM APIs with Python SDKs [CampusX]**	Using google-genai or openai SDK, system prompts, temperature, structured outputs.	Watch Video
[ ]	**Day 122**	**Streaming Token Responses in FastAPI**	FastAPI StreamingResponse, Server-Sent Events (SSE), real-time token delivery to UI.	Watch Video
[ ]	**Day 123**	**Enforcing JSON Schemas with Pydantic**	Guaranteed structured outputs from LLMs: answer, citations list, confidence score.	Watch Video
[ ]	**Day 124**	**RAG Evaluation with Ragas / TruLens**	Automated scoring of RAG outputs: measuring hallucinations without human labeling.	Watch Video
[ ]	**Day 125**	**Packaging, Testing & Deploying RAG API**	Dockerizing vector store + FastAPI, writing automated integration tests with pytest.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 19: Tool Calling & Agentic Loops (CampusX LangGraph)

Phase: Phase 5: Agentic AI, MCP & Capstone Platform | **Dates**: Mar 8 – Mar 14, 2027 (Agentic Systems Week)
Primary Goal: Move beyond single-turn RAG: build autonomous multi-step reasoning agents that use tools and maintain state.
English Resource: LangGraph Documentation: Multi-Agent Systems & State Machines | **Math**: Finite State Machines (FSM), Directed Graphs & Loop Termination Convergence
Notes & Guides: CampusX LangGraph Guide & LangGraph Official Architecture Docs
Weekly Milestone: **Build an Agent that queries your PostgreSQL database and executes actions in natural language**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 127**	**From Chains to Autonomous Agents [CampusX Agents]**	The ReAct framework (Reasoning + Acting), perception-action loop, stopping conditions.	Watch Video
[ ]	**Day 128**	**Function Calling / Tool Calling Mechanics [CampusX]**	How LLMs generate JSON function arguments, executing Python functions safely.	Watch Video
[ ]	**Day 129**	**LangGraph Foundations: State, Nodes & Edges [CampusX]**	StateGraph, TypedDict state schema, adding processing nodes, conditional branch edges.	Watch Video
[ ]	**Day 130**	**Handling Cyclic Loops & Stop Conditions**	Preventing infinite loops, maximum iteration limits, error recovery during tool execution.	Watch Video
[ ]	**Day 131**	**Multi-Turn Memory & Checkpointing**	MemorySaver, persisting conversation state to PostgreSQL, resume previous conversations.	Watch Video
[ ]	**Day 132**	**Human-in-the-Loop Approval Workflows**	Pausing agent execution before high-risk actions (e.g., executing DB writes or sending emails).	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 20: Model Context Protocol (MCP) Integration

Phase: Phase 5: Agentic AI, MCP & Capstone Platform | **Dates**: Mar 15 – Mar 21, 2027 (Model Context Protocol Week)
Primary Goal: Master Model Context Protocol (MCP), the universal standard for exposing data and tools to AI systems.
English Resource: Official MCP Docs: Building Servers, Tools, Resources & Prompts | **Math**: JSON-RPC Protocol Math: Framing, Request-Response Serialization & Transport Latency
Notes & Guides: Official Model Context Protocol (MCP) Specification & Architecture
Weekly Milestone: **Build a custom MCP Server in Python exposing backend database queries and analytics to AI clients**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 134**	**What is MCP and Why Does it Matter?**	The fragmentation of AI integrations; why MCP is the USB-C of AI development.	Watch Video
[ ]	**Day 135**	**MCP Architecture: Host, Client, Server**	STDIO transport vs Server-Sent Events (SSE); JSON-RPC message formatting.	Watch Video
[ ]	**Day 136**	**The 3 MCP Primitives: Tools, Resources, Prompts**	Tools (model-executable actions), Resources (read-only data streams), Prompts (templates).	Watch Video
[ ]	**Day 137**	**Building an MCP Server in Python**	Using mcp official Python SDK, @mcp.tool() decorator, defining typed inputs with Pydantic.	Watch Video
[ ]	**Day 138**	**Connecting AI Clients to Your MCP Server**	Configuring Claude Desktop, Cursor, and Antigravity to consume your custom MCP tools.	Watch Video
[ ]	**Day 139**	**Enterprise Security in MCP**	Input sanitization, enforcing read-only database connections, rate limiting tools.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 21: CAPSTONE FLAGSHIP PROJECT — Unified AI Knowledge Platform

Phase: Phase 5: Capstone Flagship Project | **Dates**: Mar 22 – Mar 28, 2027 (Capstone Launch Week)
Primary Goal: Unite everything learned: Backend API + PostgreSQL + Redis + Vector RAG + LangGraph Agent + Custom MCP Server.
English Resource: Full-Stack AI Architecture Guide: End-to-End Enterprise Design | **Math**: System Capacity Planning: Memory Budgets, Vector Storage Scaling & Concurrent Users
Notes & Guides: Flagship Capstone Spec: Architecture Blueprint & Deployment Guide
Weekly Milestone: **FLAGSHIP CAPSTONE LIVE: Full AI Support & Knowledge Platform deployed with video demo**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 141**	**System Design & Architecture Spec**	Client -> FastAPI -> JWT Auth -> PostgreSQL -> Redis -> Vector DB -> LLM Agent -> MCP Server.	Watch Video
[ ]	**Day 142**	**Building the Core Backend & Auth Layer**	FastAPI endpoints, user authentication, permission roles, database models.	Watch Video
[ ]	**Day 143**	**Document Ingestion & Vector Service**	Background document processing, chunking, embeddings, pgvector storage.	Watch Video
[ ]	**Day 144**	**Agentic Tool Calling & Reasoning Engine**	LangGraph state machine calling database tools, search tools, and calculating outputs.	Watch Video
[ ]	**Day 145**	**Custom MCP Server Integration**	Exposing the internal backend services through an external MCP server.	Watch Video
[ ]	**Day 146**	**Docker Compose Orchestration & Demo Video**	One-command docker compose up; recording a polished 2-minute walkthrough video.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]

Week 22: Polish, Résumé, System Design & Interview Sprint

Phase: Phase 5: Career & Interview Readiness | **Dates**: Mar 29 – Apr 4, 2027 (Career Sprint Week)
Primary Goal: Audit your portfolio, craft a high-impact AI/ML Engineer resume, prepare system design answers, and start applying.
English Resource: Gaurav Sen: System Design Playlist & Machine Learning System Design Interview Guide | **Math**: Estimation Math: Back-of-the-envelope calculations for storage, bandwidth & GPU RAM
Notes & Guides: AI/ML Engineer Resume Template & 20 Core Interview Questions Bank
Weekly Milestone: **FINAL MILESTONE: Tailored Resume, Polished GitHub Portfolio & 10 Job Applications Sent**

Done	Day	Video Title & Number	Specific Topics Covered In This Video	Direct Link
[ ]	**Day 148**	**GitHub Profile Overhaul & Portfolio Audit**	Pinned repositories, clean READMEs, badges, architecture diagrams, live demo links.	Watch Video
[ ]	**Day 149**	**1-Page AI/ML Engineer Résumé Crafting**	Focusing on quantifiable metrics: latency reduction, cache hit rates, model precision.	Watch Video
[ ]	**Day 150**	**ML System Design: Scaling Inference & RAG**	Batch vs online inference, caching embeddings, vector index partitioning, API gateways.	Watch Video
[ ]	**Day 151**	**Behavioral & Technical Question Bank**	Explaining tradeoffs: RAG vs Fine-tuning, dealing with hallucinations, data drift.	Watch Video
[ ]	**Day 152**	**Live Mock Interview Simulation**	Practicing 45-minute coding & architecture walkthrough aloud under time constraints.	Watch Video
[ ]	**Day 153**	**Job & Internship Application Sprint**	Submitting 10-15 targeted applications for AI Engineer / Backend AI roles with custom notes.	Watch Video

Weekly Pen Notes & Progress Log: _____
Hours Studied: ____/12 | Blockers/Questions: _____ | Next Week Ready: [ ]